

EXISTING COURSES:

0) Introduction to Basic Programming Concepts - Data Science Workshop

ID: DSW-000

About This Course

Target Group

Anybody did never attend a programming class or forgot almost everything...

Content

In this training we are explaining very basic programming skills. If you have never heard about if-statements, while loops or data structures / organization of data or don't know that computer programs need to define variables by content (text or numeric) and size you need to attend this class. All others can skip it, since we will explain in the other classes how to do it in Python (but not why it is needed).

- Type of variables - and why programs need to define them
- Why do we need to organize our data in different data structures?
- What are if-Statements
- How can while-loops be helpful

Prerequisites

- Laptop (if classroom training)
- EnergyAI Account (Go to: <https://energyai.siemens.cloud/> (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)

1) Python & Numpy - Data Science Workshop
ID: DSW-001

About This Course

Target Group

Any engineer who needs to analyze data or wants to program python scripts.
Content

Content

Python is a powerful tool to analyze huge amounts of data with high calculation speed. This module gives you an overview of the different data structures in Python (lists, dictionaries, etc.). We will also introduce the algebra library Numpy, which is the backbone of Python's big data options. You will learn basic commands such as loops, if statements and definitions. Our goal is to introduce you to some basic concepts. You will know enough to start yourself trial/error learning and consulting online sources.

- Intro Jupyter Notebook
- Lists, Object Localization
- Loops
- Dictionaries
- If
- Definitions
- Intro to Numpy
- ND-Arrays

Prerequisites

- Have a very basic understanding of general concepts like loops, if-statements or the need to define data types/ or attend "0) Introduction to Basic Programming Concepts - Data Science Workshop, DSW-000"
- Laptop (if classroom training)
- EnergyAI Account (Go to: <https://energyai.siemens.cloud/> (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)

2) Pandas - Data Science Workshop

Intern

ID: DSW-002

About This Course

Target Group

Any engineer who needs to analyze data or wants to program python scripts.

Content

Pandas is the part of Python used to process data. To make efficient calculations, we need to arrange data in statistical format (like for Excel Pivot-Tables or Minitab: columns for variables, rows for units, only one data type in each column). Also, we like to avoid missing values, correct data errors etc. All this data manipulation can be done very effectively with Pandas. After the training, you should be able to load, manipulate and analyze your own data in Python. Like always our goal is to introduce and practice some basic concepts. You will know enough to start your own learning by trial & error and consulting online sources.

- Intro pandas, dataframes
- Importing file
- Selection of objects
- Resample and Fill
- Calculation with dataframes
- Connecting dataframes
- Seaborn:
 - o Overview
 - o X-Axis = Continuous Data
 - o X-Axis = Categorical Data

Prerequisites

- Attend "1) Python & Numpy - Data Science Workshop, DSW-001" or read through the content on EnergyAI (shared > users > kay.toepfer@siemens.com > Data Science Workshop > Python.ipynb; Numpy.ipynb)
- Laptop (if classroom training)
- EnergyAI Account (Go to: <https://energyai.siemens.cloud/> (using Chrome, Edge or Firefox; Zscaler has to be active) and Request access to "Analyse" at least 1 week before the start of the training.)